① Asset — from the inventory

INF-01
GNSS / navigation sources

Layer Infrastructure & physical

Affected parts GNSS receiver, flight controller

Feeds geo-awareness service (SVC-05), flight authorization (SVC-03)

Evidence confidence Strong

② STRIDE — what can go wrong

S	**T**	R
I	**D**	E

S Spoofing (authenticity): a spoofed signal is accepted as genuine

T Tampering (integrity): navigation data are altered

D Denial of service (availability): jamming removes positioning

R, I, E: not assigned

③ PASTA — attack scenario

Stage focus 3–4: threat and vulnerability analysis

Vector external RF spoofing, signal interference

Weakness no cryptographic authentication of civil GNSS; single-source dependency

Scenario spoofing deviates a UAS into restricted airspace; jamming causes complete loss of positioning

Damage loss of accurate positioning

④ DREAD-S — severity

rating $x \to v(x) = 1 + 4(x-1)/9$

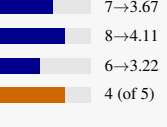


S=4: SAIL IV–V, ARA B–C (loss of separation)

$$\frac{\sum v + S}{6} = \frac{19.22 + 4}{6} = \mathbf{3.87}$$

⑤ Result — and propagation

3.87
DREAD-S, band H (High, 3.4–4.2)



chain K1: geo-awareness corrupted, authorization acts on it

Regulatory anchors EU 2021/664 Art. 15(1)(f); Part-IS IS.I.OR.200; SORA SAIL IV–V